Manipulative Language Detection in LLM-Crafted Phishing Attacks

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Abstract

This project investigates the use of transformer-based NLP models to detect manipulative language in LLM-generated phishing emails. Using the MentalManip dataset, we fine-tuned multiple models, with RoBERTa achieving the highest accuracy (0.73) for binary classification of manipulative dialogue. Further experiments targeted the "persuasion" sublabel and applied weighted loss to address class imbalance, improving recall but increasing false positives. When applied to real-world phishing emails, the models showed high sensitivity but poor precision, likely due to format mismatch and limited training data. Results highlight challenges in generalizing manipulation detection, manipulation detection is difficult even for humans.

1 Introduction

The human factor remains central in cyber attacks. The 2024 Verizon DBIR report [1] notes that 68% of breaches involve the human element, with phishing as a key contributor. With LLM tools, bad actors can now craft highly convincing phishing messages that evade traditional detection. This project investigates whether NLP models can detect manipulative language—specifically, text designed to influence actions not in the reader's best interest.

Since phishing often exploits human psychology through language, this study focuses on detecting manipulative language and whether such detection may improve defenses against phishing. Our approach first models manipulation using the "*MentalManip*" dataset, then explores its potential for phishing detection.

2 Literature

Salloum et al. [2] provide an overview of current ML and NLP methods used for phishing detection, which forms the foundational context for this project. Suhaima et al. [3] trained models like BERT on spam data, whereas our focus will be on specifically detecting manipulative language. Wang et al. [4] created a data set aiming at dialogue manipulation, which will serve as our primary training set. Al-Subaiey et al. have compiled a large corpus of emails in [5] from various datasets, under phishing-specific email body texts; this will be used for attempts to detect phishing texts.

3 Datasets

Labeled data sets focused on manipulation are rare. Most of the research has come from psychology, which provides insight into the techniques used for manipulation rather than bulk data suitable for AI model training. Most existing data sets suitable for NLP applications are concerned with hate speech and abusive language, which has been an important topic in relation to social media.

4 "MentalManip" Dataset

Wang et al. [4] introduced the "MentalManip" dataset, published on hugging face [6]. The data set is based on fictional dialogues from "The Cornell Movie Dialogs Corpus" [7] from which suitable manipulative dialogues were selected using BERT and GPT-4 models, from these, 4000 dialogues were manually selected to form the data set. The data is labeled with a detailed manipulation taxonomy in three dimensions; see Figure 1, adding applied technique and psychological vulnerability mechanism to the binary presence of whether the dialogue contains manipulation or not. Some examples of dialogues are listed in appendix E.

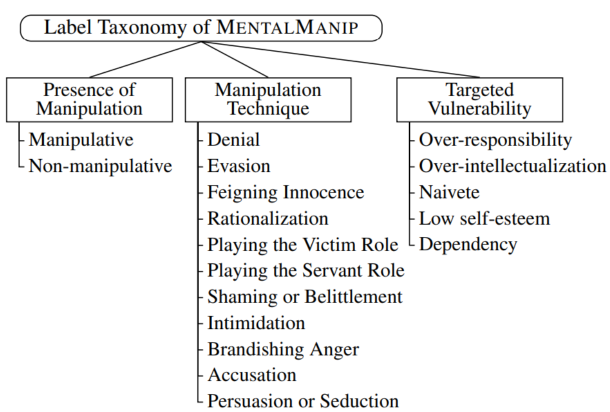


Figure 1: *Taxonomy labels in the data set*

The dataset was manually labeled through a three-step human annotation process using an adapted taxonomy (Figure 1). This resulted in two versions: one based on majority agreement ("$MentalManip_{maj}$") and one on full consensus ("$MentalManip_{con}$"). We used $MentalManip_{maj}$, which has 4000 rows, to leverage its larger size for training. Dialogues were trimmed to fit within BERT's 512-token limit. Data exploration is in Appendix A.

4.1 Labels

The dataset uses a complex labeling system based on the taxonomy in Figure 1, with uneven label distribution—'Manipulative' appears 2.4× more than 'Non-manipulative'. Some 'Technique' and 'Vulnerability' fields have multiple or missing values. See details in Appendix A.2. For phishing detection the most frequent label, 'Persuasion and Seduction', seemed most interesting (the aim is to get people to take some action on behalf of the attacker) though this creates class imbalance (4.2× more negatives than positives).

5 Baselines

Given the short embeddings (see Appendix A), basic BERT models are sufficient for the data. Due to label inconsistencies, we will focus on binary classification of 'Presence of Manipulation' as the baseline for further experimentation.

5.1 Binary with BERT and its Variants

Models trained on the 'manipulative' labels used the $MentalManip_{maj}$ dataset. They were run with similar parameters, and the accuracy at the epoch before significant overfitting[1] was recorded. The following models were investigated:

- BERT-base [8]
- RoBERTa [9]
- DistilBERT [10]
- ModernBERT [11]
- DeBERTaV3 [12]

Furthermore some "emotionally wiser" BERT derivatives exist which are pre-trained for emotion detection:

- BERTweet [13]
- EmotionBERT [14]

5.2 Baseline Results and Discussion

Baseline results are shown in Table 1. Most models overfit after a few epochs, which is typical for fine-tuning pre-trained models. Accuracy ranged from 0.70 to 0.72 with limited variation. Model details are in Appendix B.

ModernBERT ModernBERT showed no advantage, as short input lengths (Figure A.2) didn't leverage its larger capacity.

Emotionally Intelligent BERT BERTweet performed similarly to BERT-base. It is tuned for tasks like POS tagging and text classification [13], not manipulation detection. EmotionBERT [14], designed for emotional phrase classification, was also tested.

Advanced BERTs DestilBERT and DeBERTa_v3_small didn't outperform RoBERTa and required more resources. Only the smallest DeBERTa variant could be used due to hardware constraints, making these models less practical.

[1] Significant overfitting is defined as: training loss / evaluation loss < 0.6

RoBERTa RoBERTa performed best, slightly outperforming BERT-base. It was also more stable across runs, consistently achieving over 0.72 accuracy and resisting overfitting for longer.

Model	**Epoch**	**Loss T/V**	**Acc Epoch**	**Acc Final**
BERT	2	0.97	0.726	0.70
roBERTa	4	1.03	0.728	0.73
deBERTa_v3	3	0.68	0.704	0.72
DistilBERT	2	0.75	0.709	0.72
ModernBERT	2	0.81	0.718	0.72
BERTweet	2	0.97	0.705	0.70
EmotionBERT	2	1.04	0.712	0.74

Table 1: *Base Model performance comparison across different transformer architectures for binary inference on the "manipulative" column: Epochs before significant over-fitting, Training loss/Validation loss (to measure overfitting) and accuracy at epoch and final classification*

6 Fine Tuning and Experiments

Based on the results in section 5.2, we decided to fine tune the RoBERTa model with the $MentalManip_{maj}$ data set to see if we can get better results for the manipulation detection task. The MentalManip article[4] achieved an accuracy of 0.78.

6.1 LoRA/PEFT

To improve the training efficiency, LoRA was used on the RoBERTa model with parameters that will only inject lightweight, low rank trainable adaptors applied to the attention mechanism, optimize for classification task, and add regularization. This was done to get task-specific learning with minimal memory footprint and strong generalization. The final accuracy of classification was .69 slightly less than the accuracy without LoRA applied (.73).

To further improve classification efficiency, we unfroze layers 6 through11 in the RoBERTa model. Freezing these layers helps to capture interactions between words and phrases in context (6 through8) and encodes high-level semantic and task-relevant abstractions (words' meaning and why it matters)(9 through 11). We obtained the overall accuracy of .74 when only layers 10 and 11 were unfrozen.

Class	**Prec.**	**Rec.**	**F1**	**Sup.**
Non-manip.	0.52	0.23	0.32	236
Manipulative	0.74	0.91	0.82	564
Accuracy			0.71	800
Macro Avg	0.63	0.57	0.57	800
Weigt. Avg	0.67	0.71	0.67	800

Table 2: *Classification report: non-manipulative vs manipulative for the fine tuned RoBERTa model*

7 Persuasion

The 'technique' column in the data set was chosen with binary label '1' with 'persuasion' being present. The label distribution is 4.2 to 1 for the 'non-persuasion' to 'persuasion' labels, i.e. near the opposite of the data sets 'manipulative' vs 'not' when including all techniques and vulnerabilities.

7.1 Results and Discussion, Persuasion

The classification report for the 'Persuasion' label (Table 3) shows overall accuracy of 0.80. However, key metrics—precision and recall for the 'Manipulative' class—are low. Only 28% of actual manipulative cases were correctly identified (see confusion matrix in Appendix C.2b). This is likely due to the skewed label distribution.

Class	**Prec.**	**Rec.**	**F1**	**Sup.**
Non-manip.	0.85	0.92	0.88	649
Manipulative	0.46	0.28	0.35	151
Accuracy			0.80	800
Macro avg	0.65	0.60	0.61	800
Wght. avg	0.77	0.80	0.78	800

Table 3: *Classification Report for the 'Persuasion' label, Base Case un-weighted*

7.2 Weighted loss function

To address label imbalance, we applied class weights to the loss function—penalizing the majority class and boosting the minority. Under-sampling was ruled out due to the small dataset (4,000 rows), and oversampling was avoided to prevent overfitting from repeated samples. Zhang et al. [15] found loss weighting more effective than sampling when fine-tuning BERT on imbalanced data.

Weights, based on label distribution, were applied to the cross-entropy loss function [2]. We tested several schemes:

1. Inverse class distribution (Figure 2)
2. Inverse distribution with max weights capped at 4.0 and 3.0

[2] See links to notebooks in Appendix C.3

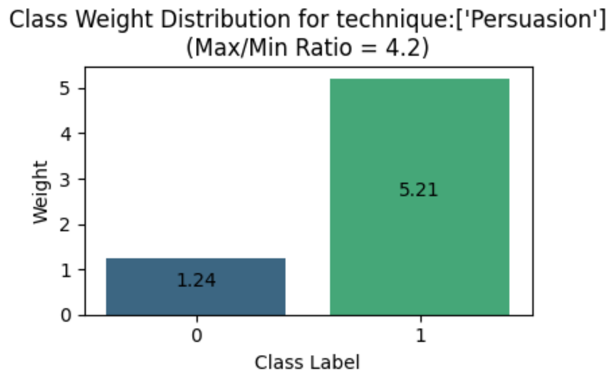
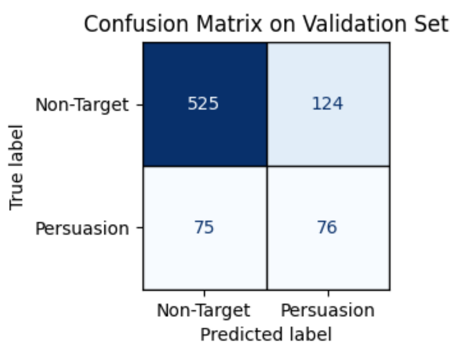


Figure 2: *Distribution of weights for the Persuasion label, no modifications to the weight distribution*



Accuracy Score: 0.75

	precision	recall	f1-score	support
Non-Target	0.88	0.81	0.84	649.0
Persuasion	0.38	0.5	0.43	151.0
macro avg	0.63	0.66	0.64	800.0
weighted avg	0.78	0.75	0.76	800.0

Figure 3: *Results for un-capped and un-normalized weighted cross-entropy loss function*

Results and Discussion, Weighted Models Option 1 (un-capped, un-normalized) performed best, as shown in Figure 3. It gave the highest weight to the minority class, improving recall for 'Manipulative' to 0.50—better than the unweighted result (Table 3)—but still only correctly identified 38% of manipulative cases, with many false positives. Option 2, with capped weights, gave lower recall but slightly better overall accuracy by favoring the majority class (Links to notebooks in in Appendix C.3). An observed feature during training, shown in Figure 4, epochs 2-4 with decreasing training loss, increasing validation loss while F1 and recall scores still show a positive trend. This is likely due to the model over fitting the majority class, however for this application we are interested in the minority class, so recall and F1 are better metrics for detecting "over fitting" of the minority class. It should be noted that multiple runs of the same model created varying results for the minority persuasion class, sometimes with no clear trend in results over epochs i.e. the model learning poorly. Some examples of dialogues vs model prediction can be found in Appendix E.

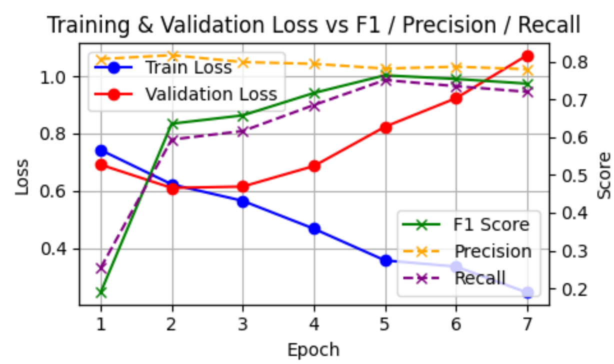


Figure 4: *Losses and Results vs epochs (Note: the model is allowed to run beyond overfit at epoch 3 fowr illustration)*

8 Phishing Detection

We further used the model with LoRA and layers 9 - 11 frozen to execute inference on Phishing email data we obtained from Kaggle. [16] This dataset included 7 different files of data. We picked two that were in English (Ling.csv and SpamAssasin.csv). They included body of the phishing email and data was labeled to indicate the binary classification between manipulative and non-manipulative.

The Ling.csv file contained 2859 records and only 458 out of them were manipulative. We got the overall accuracy of 0.29 for identifying manipulative with precision of 0.18, recall of 0.96 and F1-score of 0.30. The dataset was imbalanced with less manipulative records. The results show that our model was over-sensitive to detecting manipulative language and rarely missed identifying manipulative cases(high recall) but also flagged too many non-manipulative cases as well (low precision).

The SpamAssassin.csv file contained 5809 records and only 1718 out of them were manipulative. We got the overall accuracy of 0.43 for identifying manipulative with precision of 0.33, recall of 0.92 and F1-score of 0.49. The dataset was still moderately imbalanced with less manipulative records. The results were only slightly different from what we observed with the Ling.csv file. Our model was over-sensitive to detecting manipulative language (high recall) but struggled to identify them correctly (low precision) and overall classification ability is weak.

The data in these two files was not in dialogue form, unlike the format used during initial model training (see examples in appendices G and F). Instead, they consisted of standalone emails, which may lack the turn-taking structure or conversational cues that the model was originally optimized to interpret. Additionally, both datasets were imbalanced, with manipulative records making up a relatively small percentage of the total. These two factors—format mismatch and class imbalance—likely contributed to the model's lower accuracy. To improve performance on such data, a two-stage approach could be effective: the first model prioritizes high recall to ensure most manipulative messages are identified, while the second model focuses on improving precision by filtering out false positives. This cascaded strategy could help strike a better balance between sensitivity and specificity, ultimately improving overall reliability for real-world application.

Perhaps, a two stage model - one that focusses on high recall and second which is precision focussed can help to improve overall accuracy for these datasets.

9 Conclusion

The baseline investigations showed that applying the *MentalManip* dataset to models optimized for emotional language detection does not automatically improve accuracy. Mental manipulation is less explored in literature than hate speech and abusive language, which are key concerns in social media, where sentiment analysis is both established and evolving in NLP, using these models do not directly translate to improvement in detecting manipulation.
Investigating 'persuasion' as a means for detecting Phishing, requires some model manipulation as we are trying to detect a minority class, and the results show moderate (at best) viability for detecting persuasion.

Several factors may explain the models' poor performance on the phishing dataset. The training data is based on movie dialogues, while the phishing data consists of single emails. The dataset is also small and unbalanced. This study underscores the importance of handling class imbalance, domain adaptation, and layered model design – and that detecting manipulation is simply hard, also for humans.

References

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Individual Contributions

- **Karl-Johan Westhoff** - Drafting initial project proposal, Literature Review, Data Exploration, Baseline modeling and reporting, Experiments with Persuasion and reporting, Reporting and git infrastructure setup.
- **Neha Dhage** Finalizing project proposal, Initial data exploration and setup, Initial BERT-base model test, modeling and reporting for model Fine Tuning, Experiments with Phishing detection and reporting.
- **Both** Introduction, Conclusion and final deliverable review.

A Data Exploration

A.1 Dialogues

The 4000 dialogues in the data set are between two persons exchanging sentences. Word count statistics are shown in Figure A.1, most dialogues consist of up to 50 words per person, and the number of words uttered by each person is fairly balanced, with person 2 saying slightly more words than person 1 in the up to 50 word majority case. Figure A.2 shows the distribution of token counts for the dialogues in the data set, tokenized using BERT-base as reference. Only a minor number of dialogues exceed the BERT-base embedding size of 512 tokens.

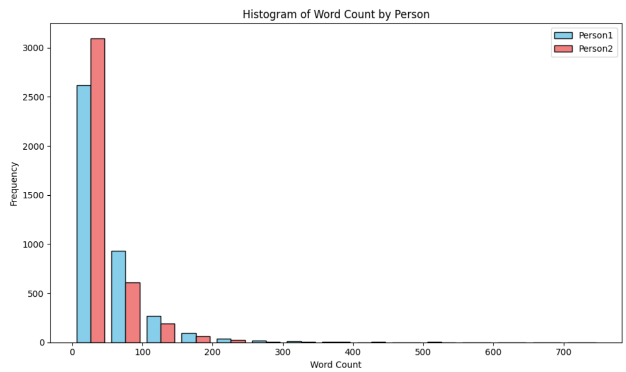


Figure A.1: *Word count statistics for the dialogues in the $MentalManip_{maj}$ data set, words uttered by each person*

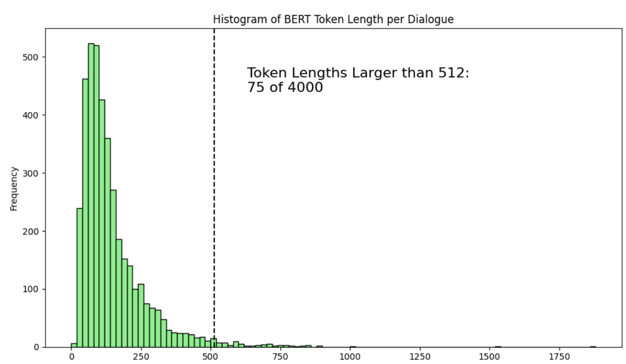


Figure A.2: *Statistics for the dialogue in the $MentalManip_{maj}$ data set, tokenized using BERT-base*

A.2 Labels

Manipulation Label The data set is not split equally between manipulation and non-manipulation, Figure A.3 shows the distribution with 2.4 times more manipulation rows than non-manipulation.

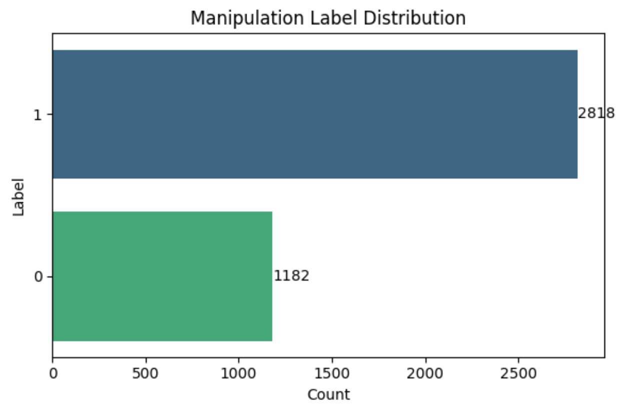


Figure A.3: *Ratio of manipulation to non-manipulation in the $MentalManip_{maj}$ Dataset*

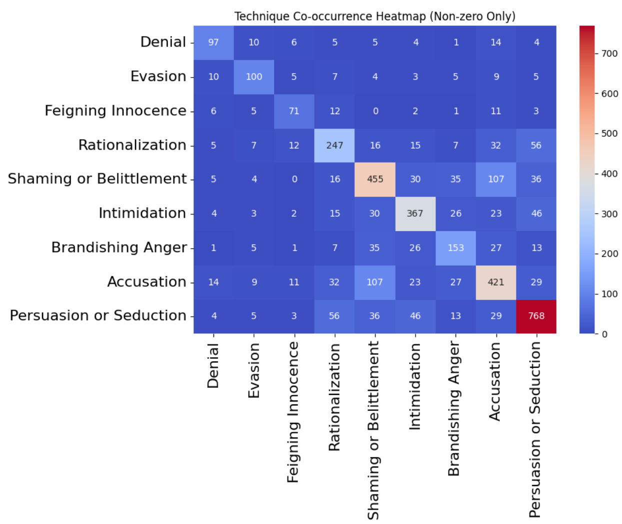


Figure A.4: *Distribution and Co-ocurrence of 'Manipulation Technique' labels*

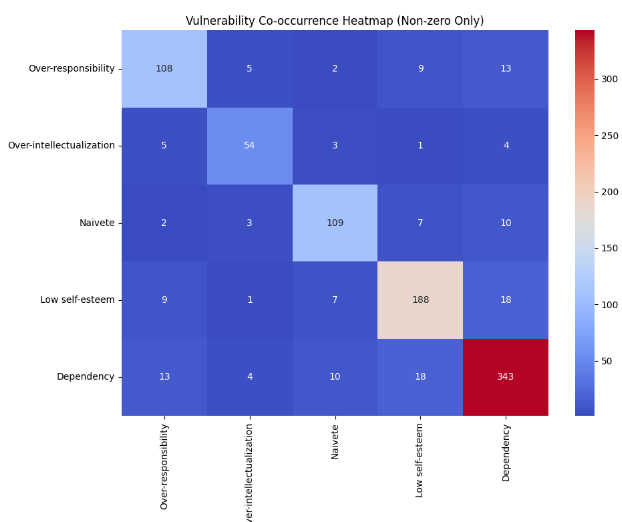


Figure A.5: *Co ocurrence of the Vulnerability column in the data $MentalManip_{maj}$ Dataset*

'Technique' and 'Vulnerability' Labels Some of the labels are missing for some of the rows with manipulation present[3], Figure A.6 shows a total of 664[4] missing labels for 'technique', we regard the technique labels as most relevant for phishing, especially the 'Persuasion or Seduction' label.

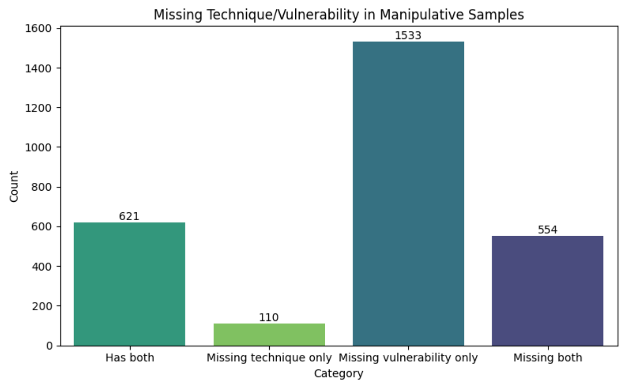


Figure A.6: *Incomplete labeling of the MentalManip Dataset*

`https://github.com/KJWesthoff/266FinalProject/blob/main/Data-Exploration_MentalManip.ipynb`

B BaseCaseModels

Notebook here:
`https://github.com/KJWesthoff/266FinalProject/tree/main/BaseCaseModels`

[3] The labels should not be populated for non-manipulation rows
[4] 110 missing technique and 554 also missing vulnerability

C 'Persuasion' Experiments

C.1 Label distribution

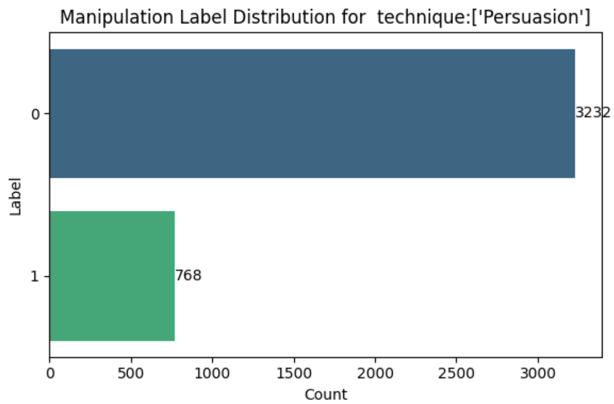


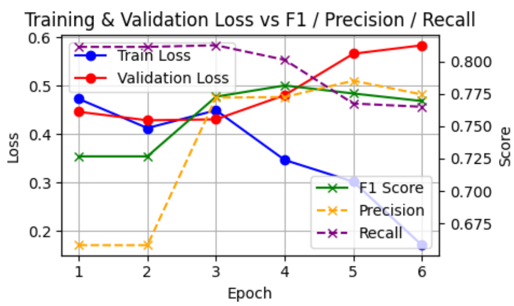
Figure C.1: *Label distribution with a 4.2 to 1 ratio for the 'not-persuation' to 'persuasion' labels in the* $MentalManip_{maj}$ *data set*

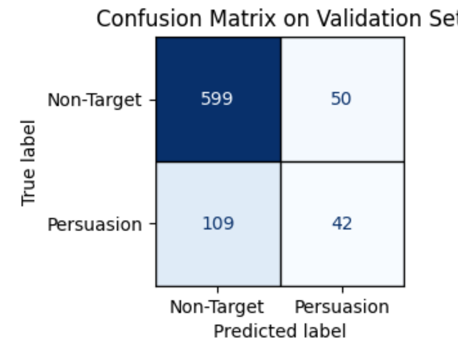
C.2 Baseline Results, Un-Weighted

Notebook here:
`https://github.com/KJWesthoff/266FinalProject/blob/main/WeightedSkew/RoBERTa_Binary_ManipDetection_PersuasionBinary_Un-Weighted.ipynb`

C.3 Baseline Results, Weighted

Notebooks here:
`https://github.com/KJWesthoff/266FinalProject/blob/main/WeightedSkew/RoBERTa_Binary_ManipDetection_PersuasionBinary_WeightedV1.ipynb`
`https://github.com/KJWesthoff/266FinalProject/blob/main/WeightedSkew/RoBERTa_Binary_ManipDetection_PersuasionBinary_WeightedV2.ipynb`
`https://github.com/KJWesthoff/266FinalProject/blob/main/WeightedSkew/RoBERTa_Binary_ManipDetection_PersuasionBinary_WeightedV3.ipynb`





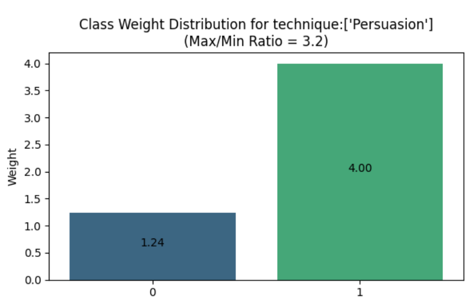
Accuracy Score: 0.8

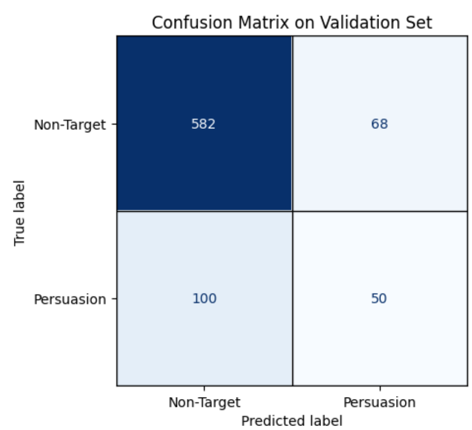
	precision	recall	f1-score	support
Non-Target	0.85	0.92	0.88	649.0
Persuasion	0.46	0.28	0.35	151.0
macro avg	0.65	0.6	0.61	800.0
weighted avg	0.77	0.8	0.78	800.0

(a) *Losses and Results vs epochs for the un-weighted model*

(b) *Confusion matrix and classification report for the un weighted Persuasion label base case model*

Figure C.2: *Results for the un weighted persuasion label base case model*





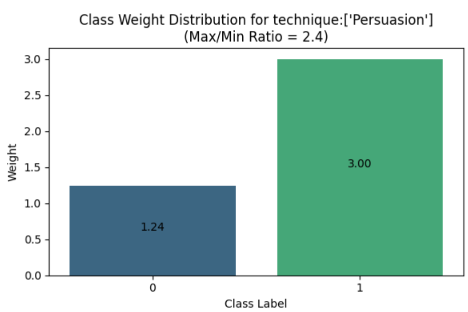
Accuracy Score: 0.79

	precision	recall	f1-score	support
Non-Target	0.85	0.9	0.87	650.0
Persuasion	0.42	0.33	0.37	150.0
macro avg	0.64	0.61	0.62	800.0
weighted avg	0.77	0.79	0.78	800.0

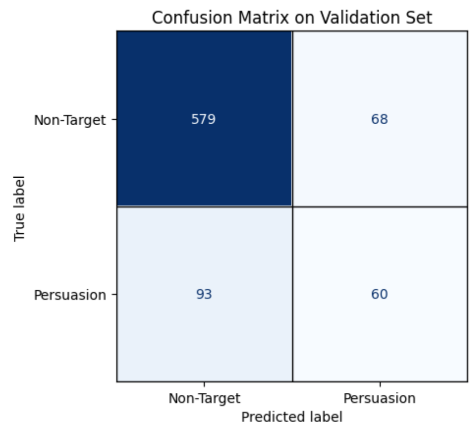
(a) *Distribution of weights for the Persuasion label*

(b) *Confusion matrix and classification report for the Persuasion label base case model*

Figure C.3: *Results for cross entropy weight capped at 4.0*



(a) *Distribution of weights for the Persuasion label*



Accuracy Score: 0.8

	precision	recall	f1-score	support
Non-Target	0.86	0.89	0.88	647.0
Persuasion	0.47	0.39	0.43	153.0
macro avg	0.67	0.64	0.65	800.0
weighted avg	0.79	0.8	0.79	800.0

(b) *Confusion matrix and classification report for the Persuasion label base case model*

Figure C.4: *Results for cross entropy weight capped at 3.0*

D Fine Tuning Notebooks

Notebooks here:
https://github.com/KJWesthoff/266FinalProject/blob/main/FineTuning/RoBERTa_Binary_ManipDetection_LoRA_Freeze%20layer)_10-11_Integrated.ipynb
https://github.com/KJWesthoff/266FinalProject/blob/main/FineTuning/RoBERTa_Binary_ManipDetection_LoRA_Freeze%20layer)_6-through-11_Integrated.ipynb
https://github.com/KJWesthoff/266FinalProject/blob/main/FineTuning/RoBERTa_Binary_ManipDetection_LoRA_Freeze%20layer)_8-9-10-11_Integrated.ipynb
https://github.com/KJWesthoff/266FinalProject/blob/main/FineTuning/RoBERTa_Binary_ManipDetection_LoRA_Freeze%20layer)_8-9_Integrated.ipynb
https://github.com/KJWesthoff/266FinalProject/blob/main/FineTuning/RoBERTa_Binary_ManipDetection_LoRA_Integrated%20(1).ipynb
https://github.com/KJWesthoff/266FinalProject/blob/main/FineTuning/RoBERTa_Binary_ManipDetection_PEFT.ipynb
https://github.com/KJWesthoff/266FinalProject/blob/main/FineTuning/RoBERTa_Binary_ManipDetection_PEFT_rev2.ipynb

E Dialogue examples, Persuasion Detection

Code Here:
https://github.com/KJWesthoff/266FinalProject/blob/main/WeightedSkew/RoBERTa_Binary_ManipDetection_PersuasionBinary_WeightedV1d.ipynb

E.1 True Positives:

Person1: You're making it too easy.
Person2: You got time on your side. Pretty soon they'll be missed and we'll have the law up our ass.
Person1: They saw you kill the driver.
Person2: You're up on your details, aren't you?
Person1: You can rely on them to keep quiet because this is undeclared money that could land Jack there in federal prison. He can't afford for you to get caught and have this briefcase appear as evidence.
Person2: Keep talking.

Person1: Drink this! It will dull your pain.
Person2: It will numb my wits, and I must have them all. If I'm senseless, or if I wail, then Longshanks will have broken me.
Person1: I can't bear the thought of your torture. Take it!

E.2 False Negatives:

Person1: She's working as fast as she can, Icarus. It will be ready soon.
Person2: It's ready now, I know it is.
Person1: She says it's not.
Person2: She's lying. She lost the first one on purpose.
Person1: She did not. The mouse ran down the drain.
Person2: She let it escape because she wants me to die.
Person1: Don't be a child, Icarus. She is just another scientist and like all scientists, she doesn't care about anything outside the

Person1: Why did you have LUH come here?
Person2: Why are you so concerned?
Person1: What's going on?
Person2: I want you for my roommate.
Person1: Where's LUH?
Person2: It will be good for both of us. I've got it all arranged.

E.3 False Positives:

Person1: Yes! You own the Coffer of Shadow. Nothing can withstand its power.
Person2: I've been saving it. For the right moment.
Person1: That moment is now! What good is a sword unless it be unsheathed? Use it, and no one will dare oppose you again. No one.

Person1: Here and there. Around.
Person2: Uh-huh. One of those cozy bed and breakfast places, probably.
Person1: Yeah, that's right.
Person2: Except that there's no bed, is there? And no breakfast either.
Person1: The material world is an illusion. It doesn't matter if they're there or not. The world is in my head.
Person2: But your body is in the world, isn't it? If someone offered you a place to stay, you wouldn't necessarily refuse, would you?
Person1:

E.4 True Negatives:

Person1: I don't know who you think you're talking to! I ain't some whore you brought here! I've been trying to be your friend and you treat me like shit!
Person2: Be a friend. Leave.
Person1: You got no manners and you never tell the truth! You're nothin' special. And if you ask me, you got no chance at all of being an officer!

Person1: Oh, Rufus!
Person2: All I can offer you is a Rufus over your head.
Person1: Oh, Your Excellency, I don't know what to say.
Person2: I wouldn't know what to say either if I was in your place. Maybe you can suggest something.

F Phishing Detection Attempts Text, Ling

Code Here:
https://github.com/KJWesthoff/266FinalProject/blob/main/FineTuning/SpamAssasin/inferenceKJ_edit1_PostProc.ipynb

F.1 True Positives:

Mail-1: greetings , this email is about fighting spam , and winning . > > please forward to your friends , family and co-workers to people that want to get spam out of their email : read this , you ' ll like what i am doing . . . . :) to people that want to flame me for whatever reason : do n't be an egomaniac , i have a right to free speech , and the right to say it to a lot of people , so buzz off . - - - - - first let me tell you who i am and what i do . my name is robert young , i

Mail-1: just released . . . 27 , 000 , 000 email addresses ! plus 12 bonuses . . . including free bulk e . mail software . these addresses are less than 21 days old . earn insane profits with the right formula if you have a product , service , or message that you would like to get out to thousands , hundreds of thousands , or even millions of people , you have several options . traditional methods include print advertising , direct mail , radio , and television advertising . they are all effective , but they all have two catches : they ' re expensive and time consuming . not only that , you only get one shot at making your message

F.2 False Negatives:

Mail-1: discover your family history - rated " cool site of the week " come visit our website at , http : / / www . vpm . com / hallofnames / index . htm free search = 09 do you know who your ancestors are and what they did ? do you know when your surname first appeared ? are you curious about where your family roots originate ? now you can fill in the missing pieces of this puzzle . join the satisfied multitudes who have discovered their complete family = surname history . all nationalities . it 's easy . just key your last name into our online = index , and in seconds we will

Mail-1: discover your family history come visit our website at , http : / / www . traceit . com free search do you know who your ancestors are and what they did ? do you know when your surname first appeared ? are you curious about where your family roots originate ? now you can fill in the missing pieces of this puzzle . join the satisfied multitudes who have discovered their complete family surname history . all nationalities . it 's easy . just key your last name into our online index , and in seconds we will tell you it 's origin and much more . see if we ' ve researched your complete family name history during

F.3 False Positives:

Mail-1: istituto mitteleuropeo di cultura - mitteleuropaeisches kulturinstitut bisca-98 bolzano international school in cognitive analysis unfolding perceptual continua bolzano , september 7-11 recent problems raised by cognitive science , such as the perception of forms , the recognition of natural languages , the problems of common sense , na 239ve physics , and consequently the need for direct and non-propositional reference to the objects of experience - as cited , for example by scientists working in robotics - has proposed new areas of inquiry for psychophysics .

Mail-1: hello , does anybody happen to know of languages (other than albanian and macedonian) which display obligatory clitic doubling of (either accusative or dative) dps ? any reference will be appreciated . please , write to me directly at : dalina . kallulli @ avh . unit . no thank you ! dalina kallulli

F.4 True Negatives:

Mail-1: i would like to place a large bet that chomsky is by far the most cited linguist in the postings to the linguist list . . .

Mail-1: forgive me if i missed this , but in the earlier discussion of chomsky 's elevated position in the citations index , did anyone work out how many of the citations were for his linguistic works and how many were for his political works . i ask this out of general interest , and also because a colleague of mine is interested to know the extent of his stature in academic political studies . comments on the latter could be directed directly to me .

G Phishing Detection Attempts Text, SpamAssassin

Code Here:
https://github.com/KJWesthoff/266FinalProject/blob/main/FineTuning/SpamAssasin/inferenceKJ_edit1_PostProc.ipynb

G.1 True Positives:

Mail-1: Dear Consumers, Increase your Business Sales!
How??
By targeting millions of buyers via e-mail !!
25 MILLION EMAILS + Bulk Mailing Software For Only $150.00
super low price! ACT NOW !!!
Our Fresh Addresses Will Bring You
Incredible Results!
If you REALLY want to get the word out regarding
your services or products, Bulk Email is the BEST
way to do so, PERIOD! Advertising in newsgroups is
good but you're competing with hundreds even TH

Mail-1: Make 20
Investor Alert:
Is the stock market roller coaster making you worried? Join the flight to quality. Make 20
Earn over 2Receivable Acquisitions-prepaid monthly in advance!!!
You heard right-over 2
Discover what banks have been doing for decades.
Harness the power & liquidity of fully secured accounts receivable acquisitions.
Look into our safe haven with a return of 20

G.2 False Negatives:

Mail-1: ATTN:SIR/MADAN
STRICTLY CONFIDENTIAL.
I am pleased to introduce myself to you.My name is Mr. Manuel Oko a native of South Africa and a senior employee of mines and natural resources department currently on a trainning course in

Mail-1: -1715101671.1022782885778.JavaMail.root.smtp4 Content-Type: text/plain; charset=iso-8859-1 Content-Transfer-Encoding: 7bit O'Connor quickly moved his cursor down the left side of the web page. Then, showing a flash of his silky skills, he cut across to the right hand side and clicked on Top Ten Special Offers. He zipped past the Irish Chicken Breast Fillets at 25

G.3 False Positives:

Mail-1: In a message dated Wed, 7 Aug 2002 5:17:55 PM Eastern Standard Time, beberg@mithral.com writes:
> > > On Wed, 7 Aug 2002, Robert Harley wrote: > > > Hmm... now I have an asymptotically faster precomputation algorithm > > Woo, faster is better... But what have you done about all those patents that > prevent anyone from going near ECC in the first place, > hmmmmmm? > > - Adam L. "Duncan"

Mail-1: » > Well, I don't really find it consistent at all to use an rpm package » > built against something that wasn't installed through rpm :- » » Following that reasoning, I've been installing all my custom-built » kernels through rpm recently. I find it annoying, though, that » alsa-kernel, and similar packages, will only build for the currently » running kernel. » » So I've attached a patch to specify an alternate kernel by setting the » "TARGET_KERNEL" environment variable before running rpmbuild. You

G.4 True Negatives:

Mail-1: CAN-2002-0384 http://cve.mitre.org/cgi-bin/cvename.cgi?name=CAN-2002-0384 RHSA-2002-098 http://rhn.redhat.com/errata/RHSA-2002-098.html Gaim Changelog http://gaim.sourceforge.net/ChangeLog
Gaim is an instant messaging client based on the published TOC protocol from AOL. Versions of Gaim prior to 0.58 contain a buffer overflow in the Jabber plug

Mail-1: Hi All,
I have a question which is a bit tricky and was wondering of anyone has come across this problem before or could point me in the right direction.
I am involved in porting a SCO unix application to Linux, and we have encountered a problem with the way semaphores are being handled. The application uses mulitple processes to run application code with the main process known as the bsh which controls all i/o be it screen, or file i/o, syncronisation is handled via semaphores.